NovoTech MACHINE TOOLS Inc.

PRODUCT CATALOGUE 2026

CSM · MADE IN TAIWAN · SOLD & SERVICED IN CANADA BY NOVOTECH

# ELECTRIC TUBE BENDERS AND WIRE BENDERS

## CSM EMB / EMR AND MRE2-RBE ALL-ELECTRIC BENDERS · MBE / MRE AND TBRE / TMRE / TSRE / TDRE ELECTRIC HYBRIDS · WB CNC WIRE BENDERS · Ø5/32″ WIRE TO Ø6″ TUBE







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CSM ELECTRIC TUBE BENDERS

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ABOUT NOVOTECH

# YOUR CANADIAN SOURCE FOR CSM ELECTRIC TUBE BENDERS

Sales, installation, parts and technical support across Canada.

**NovoTech Machine Tools Inc.** supplies metalworking and fabrication machinery to job shops, fabricators, maintenance departments and manufacturers across Canada from our base in Mississauga, Ontario. Every machine in this catalogue is sold and supported locally by NovoTech — with spare parts and technical support readily available in Canada.

**INCLUDED WITH EVERY NOVOTECH MACHINE**

- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

CSM · TAINAN, TAIWAN

# SERVO-DRIVEN TUBE BENDERS, FROM Ø15/32" BRAKE LINE TO Ø6" TUBE

CSM's electric benders put a servo motor on the bend arm — through CSM's patented transmission gears, to ±0.01° — and on the carriage, collet and tool shift. The EMB / EMR and MRE2-RBE are all-electric, with no hydraulic unit at all; the EMB / EMR switch between left- and right-hand bending within one part. The MBE / MRE and the TBRE / TMRE / TSRE / TDRE are electric hybrids with hydraulic clamp, pressure die and mandrel, one to three stacked tools, roll-bending heads and servo rear boosters, from Ø15/32" fuel line to Ø6". The WB 10-A and WB 14-A bend wire from the coil in 3D at up to 1,000°/s.

Chiao Sheng Machinery (CSM) has built tube benders in Tainan, Taiwan, since 1982 and released its first CNC bender in 1989; the company works from two plants in the Southern Taiwan Science Park, holds 24 international design patents, has been ISO 9001 certified since 2002 and CE-certifies every machine. NovoTech sells, installs and supports CSM machines in Canada from Mississauga, Ontario — layout for your building, installation, operator training, a CSA-approved panel, a 1-year warranty and parts and technical support in Canada.

15 EMB – 75 EMR · 50 – 100 MRE2-RBE · 10 MODELS

# EMB / EMR AND MRE2-RBE ALL-ELECTRIC CNC TUBE BENDERS, Ø19/32" TO Ø4"



**MODEL** EMB · CSM 45 EMR ALL-ELECTRIC CNC TUBE BENDER WITH LEFT- AND RIGHT-HAND BENDING HEAD



CSM 80 MRE2-RBE all-electric CNC tube bender with servo rear booster



15 EMB all-electric CNC bender for brake and fuel line to Ø19/32"

## All-electric · left- and right-hand bending · Ø19/32" – Ø4" · 3 + 3 stacked tools

CSM's all-electric CNC benders: every axis — bend arm, carriage, collet, clamp and pressure die, side booster, tool shift and mandrel — is a servo motor, so there is no hydraulic unit, no oil, no warm-up and about half the energy of a hydraulic bender, with ±0.01° bend-arm accuracy. The EMB / EMR work-head rotates to bend left- and right-hand in the same part — brake and fuel lines, air-conditioning and oil tube on the 15 and 20, furniture and health-care tube on the 30 – 45, automotive and transportation tube on the 75 — with 3 + 3 stacked tools and a roll-bending head. The MRE2-RBE adds a servo rear booster for Ø2" to Ø4" tube in thin walls and tight radii.

### STANDARD FEATURES

- ✓ All axes by servo motor: bend arm, carriage, collet, clamp die (EMB / EMR by pneumatic), pressure die, side booster, tool shift up-down and left-right
- ✓ Work-head rotation by servo motor — converts between left- and right-hand bending within one part (EMB / EMR)
- ✓ Mandrel by servo motor with automatic lubrication and anticipated retraction
- ✓ Servo rear booster on the MRE2-RBE and 75 EMR
- ✓ Industrial-PC control with touch-screen: bending simulation, 3D preview and interference detection; reads IGS / STP files
- ✓ Programmable speed for every servo axis; USB port for data transfer; manual / automatic mode; inch / metric; XYZ / YBC axes
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

### OPTIONAL EQUIPMENT

- + Automatic tube loader and unloader; bundle loading
- + Rear booster (servo RBE / hydraulic RBF, RBH) for thin-wall and tight-radius bending
- + Roll-bending head for large radii
- + Integrated circular saw, end-forming station and robot transfer — automated bending cells
- + Tooling: bend dies, clamp dies, pressure dies, wiper dies and mandrels to your tube and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS

# EMB / EMR AND MRE2-RBE ALL-ELECTRIC CNC TUBE BENDERS, Ø19/32" TO Ø4"

## EMB / EMR

All-electric CNC left- and right-hand benders

| Model | 15 EMB | 20 EMB | 20 EMR | 32 MBRE | 30 EMR | 45 EMR | 75 EMR |
|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø19/32" (15 mm) × 16 ga (1.5 mm) | Ø13/16" (20 mm) × 16 ga (1.5 mm) | Ø13/16" (20 mm) × 16 ga (1.5 mm) | Ø1-1/4" (32 mm) × 15 ga (1.8 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-3/4" (45 mm) × 16 ga (1.6 mm) | Ø3" (76.2 mm) × 14 ga (2 mm) |
| **Centreline radius (CLR)** | 0.4 – 2.2" (10 – 55 mm) | 0.4 – 3.1" (10 – 80 mm) | 0.4 – 3.1" (10 – 80 mm) | 1.0 – 5.0" (25 – 128 mm) | 0.6 – 3.5" (15 – 90 mm) | 1.2 – 5.9" (30 – 150 mm) | 1.6 – 9.1" (40 – 230 mm) |
| **Mandrel working length** | 3.3 ft · 39" (1,000 mm) | 4.9 ft · 59" (1,500 mm) | 4.9 ft · 59" (1,500 mm) | 11.5 ft · 138" (3,500 mm) | 9.2 ft · 110" (2,800 mm) | 9.8 ft · 118" (3,000 mm) | 9.8 ft · 118" (3,000 mm) |
| **Tooling capacity**<br>stacked dies / radii | 3+3 | 3+3 | 3+3 | 3+3 | 3+3 | 3+3 | 3+3 |
| **Roll bending** | – | – | Yes | Yes | Yes | Yes | Yes |
| **Rear booster** | – | – | – | – | – | – | Servo |

## MRE2-RBE

All-electric CNC benders with servo rear booster

| Model | 50 MRE2-RBE | 80 MRE2-RBE | 100 MRE2-RBE |
|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø2" (50.8 mm) × 15 ga (1.8 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø4" (100 mm) × 10 ga (3.6 mm) |
| **Centreline radius (CLR)** | 1.2 – 5.9" (30 – 150 mm) | 1.6 – 9.1" (40 – 230 mm) | 1.2 – 11.8" (30 – 300 mm) |
| **Mandrel working length** | 8.2 ft · 98" (2,500 mm) | 9.8 ft · 118" (3,000 mm) | 9.8 ft · 118" (3,000 mm) |
| **Tooling capacity**<br>stacked dies / radii | 3 | 3 | 3 |
| **Roll bending** | Yes | Yes | Yes |
| **Rear booster** | Servo | Servo | Servo |

Capacities are the manufacturer's ratings for round mild-steel tube, outside diameter × wall; CLR = centreline radius of the bend die. Mandrel length is the maximum tube length fed with the standard mandrel rod (the working length). Tooling capacity is the number of stacked bend dies / radii; "roll bending" models carry a roll-bending head for large radii; a rear booster (RBE servo, RBF / RBH / CBF hydraulic) pushes the tube from behind for thin-wall and tight-radius bends. Three-phase machines are supplied for 600 V with a CSA-approved panel.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

### INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

18 MBE – 150 MRE-RBE · 24 MODELS

# MBE / MRE ELECTRIC-HYBRID CNC TUBE BENDERS, Ø11/16" TO Ø6"



**MODEL** MBE · CSM MBE ELECTRIC-HYBRID CNC TUBE BENDER WITH THREE STACKED TOOLS



Stainless exhaust manifold bent on the 80 – 150 MBE / MRE



HVAC and fuel-line tube bent on the 18 – 25 MBE

## Electric hybrid · Ø11/16" – Ø6" · 3 stacked tools · clockwise and counter-clockwise

The electric-hybrid bender: servo motors on the bend arm, carriage, collet rotation and tool shift for accuracy and speed, hydraulics on the clamp die, pressure die, side booster and mandrel for force — the combination most production shops choose. Twenty-four models from Ø11/16" to Ø6" in three stacked tools; the MRE adds a roll-bending head for large radii and the -RBE versions a servo rear booster. The work-head travels on linear guides, so the same machine bends clockwise and counter-clockwise.

### STANDARD FEATURES

- ✓ Bend arm (C-axis) by servo motor through CSM's patented transmission gears — ±0.01° tolerance
- ✓ Carriage (Y-axis) by servo motor with programmable speed; collet (B-axis) rotation by servo motor
- ✓ Clamp die, pressure die and side booster by hydraulics; tool / radii shift up-down and left-right by servo motor
- ✓ Mandrel by hydraulics with automatic lubrication and anticipated retraction
- ✓ Work-head on linear guides — bends clockwise and counter-clockwise on one machine (MBE / MRE)
- ✓ Roll-bending head on the MRE; servo rear booster on the -RBE models
- ✓ Industrial-PC control with touch-screen: bending simulation, 3D preview and interference detection; reads IGS / STP files
- ✓ Programmable speed for every servo axis; USB port for data transfer; manual / automatic mode; inch / metric; XYZ / YBC axes
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

### OPTIONAL EQUIPMENT

- + Automatic tube loader and unloader; bundle loading
- + Rear booster (servo RBE / hydraulic RBF, RBH) for thin-wall and tight-radius bending
- + Roll-bending head for large radii
- + Integrated circular saw, end-forming station and robot transfer — automated bending cells
- + Tooling: bend dies, clamp dies, pressure dies, wiper dies and mandrels to your tube and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS

# MBE / MRE ELECTRIC-HYBRID CNC TUBE BENDERS, Ø11/16" TO Ø6"

## MBE

Electric-hybrid CNC benders

| Model | 18 MBE | 25 MBE | 30 MBE | 38 MBE | 50 MBE | 65 MBE | 80 MBE | 100 MBE | 65 MBE-RBE | 80 MBE-RBE | 100 MBE-RBE | 130 MBE-RBE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø11/16" (18 mm) × 16 ga (1.6 mm) | Ø1" (25.4 mm) × 16 ga (1.5 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-1/2" (38.1 mm) × 15 ga (1.8 mm) | Ø2" (50.8 mm) × 14 ga (2 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø4" (100 mm) × 11 ga (3 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.8 mm) | Ø4" (100 mm) × 10 ga (3.6 mm) | Ø5" (127 mm) × 9 ga (3.8 mm) |
| **Centreline radius (CLR)** | 0.4 – 3.1" (10 – 80 mm) | 0.6 – 4.7" (15 – 120 mm) | 0.6 – 4.7" (15 – 120 mm) | 1.0 – 5.9" (25 – 150 mm) | 1.2 – 8.7" (30 – 220 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 3.1 – 15.7" (80 – 400 mm) |
| **Mandrel working length** | 3.9 ft · 47" (1,200 mm) | 4.9 ft · 59" (1,500 mm) | 7.2 ft · 87" (2,200 mm) | 7.2 ft · 87" (2,200 mm) | 8.2 ft · 98" (2,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 13.1 ft · 157" (4,000 mm) |
| **Tooling capacity**<br>stacked dies / radii | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| **Roll bending** | – | – | – | – | – | – | – | – | – | – | – | – |
| **Rear booster** | – | – | – | – | – | – | – | – | Servo | Servo | Servo | Servo |

## MRE

Electric-hybrid CNC benders with roll-bending head

| Model | 18 MRE | 25 MRE | 30 MRE | 38 MRE | 50 MRE | 65 MRE | 80 MRE | 65 MRE-RBE | 80 MRE-RBE | 100 MRE-RBE | 130 MRE-RBE | 150 MRE-RBE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø11/16" (18 mm) × 16 ga (1.6 mm) | Ø1" (25.4 mm) × 16 ga (1.5 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-1/2" (38 mm) × 15 ga (1.8 mm) | Ø2" (50.8 mm) × 14 ga (2 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.8 mm) | Ø4" (100 mm) × 10 ga (3.6 mm) | Ø5" (127 mm) × 9 ga (3.8 mm) | Ø6" (152.4 mm) × 11 ga (3 mm) |
| **Centreline radius (CLR)** | 0.4 – 3.1" (10 – 80 mm) | 0.6 – 4.7" (15 – 120 mm) | 0.6 – 4.7" (15 – 120 mm) | 1.0 – 5.9" (25 – 150 mm) | 1.2 – 8.7" (30 – 220 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 3.1 – 15.7" (80 – 400 mm) | 3.5 – 15.7" (90 – 400 mm) |
| **Mandrel working length** | 3.9 ft · 47" (1,200 mm) | 4.9 ft · 59" (1,500 mm) | 7.2 ft · 87" (2,200 mm) | 7.2 ft · 87" (2,200 mm) | 8.2 ft · 98" (2,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 13.1 ft · 157" (4,000 mm) | 13.1 ft · 157" (4,000 mm) |
| **Tooling capacity**<br>stacked dies / radii | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| **Roll bending** | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Rear booster** | – | – | – | – | – | – | – | Servo | Servo | Servo | Servo | Servo |

Capacities are the manufacturer's ratings for round mild-steel tube, outside diameter × wall; CLR = centreline radius of the bend die. Mandrel length is the maximum tube length fed with the standard mandrel rod (the working length). Tooling capacity is the number of stacked bend dies / radii; "roll bending" models carry a roll-bending head for large radii; a rear booster (RBE servo, RBF / RBH / CBF hydraulic) pushes the tube from behind for thin-wall and tight-radius bends. Three-phase machines are supplied for 600 V with a CSA-approved panel.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

**INCLUDED WITH EVERY NOVOTECH MACHINE**

- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
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  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

25 TBRE – 130 TBRE-RBE · 25 – 100 TMRE-RBE · 18 TSRE – 100 TDRE-RBE · 42 MODELS

# TBRE / TMRE AND TSRE / TDRE ELECTRIC-HYBRID CNC TUBE BENDERS, Ø15/32" TO Ø5"



**MODEL** TBRE · CSM TMRE ELECTRIC-HYBRID CNC TUBE BENDER WITH THREE STACKED TOOLS AND ROLL-BENDING HEAD



CSM 65 TDRE electric-hybrid CNC tube bender with two stacked tools



Serpentine HVAC tube bent on the 18 – 25 TBRE

## Electric hybrid · Ø15/32" – Ø5" · 1 – 3 stacked tools · roll bending on TBRE / TMRE

The T series: the same servo bend arm, carriage and collet as the MBE on a fixed work-head with two (TBRE, TDRE) or three (TMRE) stacked tools and hydraulic tool shift up-down — the economical electric-hybrid bender for a shop that bends one direction. The TBRE / TMRE carry a roll-bending head and reach Ø5"; the TSRE (one tool) and TDRE (two tools) start at Ø15/32" for HVAC and fuel line and go to Ø4"; -RBE versions add the servo rear booster.

### STANDARD FEATURES

- ✓ Bend arm (C-axis) by servo motor through CSM's patented transmission gears — ±0.01° tolerance
- ✓ Carriage (Y-axis) by servo motor with programmable speed; collet (B-axis) rotation by servo motor
- ✓ Clamp die, pressure die and side booster by hydraulics; tool shift up-down by hydraulics, left-right by servo motor
- ✓ Mandrel by hydraulics with automatic lubrication and anticipated retraction
- ✓ Roll-bending head on the TBRE / TMRE; servo rear booster on the -RBE models
- ✓ Industrial-PC control with touch-screen: bending simulation, 3D preview and interference detection; reads IGS / STP files
- ✓ Programmable speed for every servo axis; USB port for data transfer; manual / automatic mode; inch / metric; XYZ / YBC axes
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

### OPTIONAL EQUIPMENT

- + Automatic tube loader and unloader; bundle loading
- + Rear booster (servo RBE / hydraulic RBF, RBH) for thin-wall and tight-radius bending
- + Roll-bending head for large radii
- + Integrated circular saw, end-forming station and robot transfer — automated bending cells
- + Tooling: bend dies, clamp dies, pressure dies, wiper dies and mandrels to your tube and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS · 1 / 2

# TBRE / TMRE AND TSRE / TDRE ELECTRIC-HYBRID CNC TUBE BENDERS, Ø15/32" TO Ø5"

## TBRE

Two stacked tools, roll bending

| Model | 25 TBRE | 30 TBRE | 38 TBRE | 50 TBRE | 65 TBRE | 80 TBRE | 100 TBRE | 65 TBRE-RBE | 80 TBRE-RBE | 100 TBRE-RBE | 130 TBRE-RBE |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø1" (25.4 mm) × 16 ga (1.5 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-1/2" (38.1 mm) × 15 ga (1.8 mm) | Ø2" (50.8 mm) × 14 ga (2 mm) | Ø2-1/2" (65 mm) × 14 ga (2 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø4" (100 mm) × 11 ga (3 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.8 mm) | Ø4" (100 mm) × 10 ga (3.6 mm) | Ø5" (127 mm) × 9 ga (3.8 mm) |
| **Centreline radius (CLR)** | 0.6 – 4.7" (15 – 120 mm) | 0.6 – 4.7" (15 – 120 mm) | 1.0 – 5.9" (25 – 150 mm) | 1.2 – 8.7" (30 – 220 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 3.1 – 15.7" (80 – 400 mm) |
| **Mandrel working length** | 4.9 ft · 59" (1,500 mm) | 7.2 ft · 87" (2,200 mm) | 7.2 ft · 87" (2,200 mm) | 8.2 ft · 98" (2,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 13.1 ft · 157" (4,000 mm) |
| **Tooling capacity**<br>stacked dies / radii | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **Roll bending** | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Rear booster** | – | – | – | – | – | – | – | Servo | Servo | Servo | Servo |

## TMRE

Three stacked tools, roll bending

| Model | 25 TMRE | 30 TMRE | 38 TMRE | 50 TMRE | 65 TMRE | 80 TMRE | 100 TMRE | 65 TMRE-RBE | 80 TMRE-RBE | 100 TMRE-RBE |
|---|---|---|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø1" (25.4 mm) × 16 ga (1.5 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-1/2" (38.1 mm) × 15 ga (1.8 mm) | Ø2" (50.8 mm) × 14 ga (2 mm) | Ø2-1/2" (65 mm) × 14 ga (2 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø4" (100 mm) × 11 ga (3 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.8 mm) | Ø4" (100 mm) × 10 ga (3.6 mm) |
| **Centreline radius (CLR)** | 0.6 – 4.7" (15 – 120 mm) | 0.6 – 4.7" (15 – 120 mm) | 1.0 – 5.9" (25 – 150 mm) | 1.2 – 8.7" (30 – 220 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) |
| **Mandrel working length** | 4.9 ft · 59" (1,500 mm) | 7.2 ft · 87" (2,200 mm) | 7.2 ft · 87" (2,200 mm) | 8.2 ft · 98" (2,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) |
| **Tooling capacity**<br>stacked dies / radii | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| **Roll bending** | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Rear booster** | – | – | – | – | – | – | – | Servo | Servo | Servo |

TECHNICAL SPECIFICATIONS · 2 / 2

# TBRE / TMRE AND TSRE / TDRE ELECTRIC-HYBRID CNC TUBE BENDERS, Ø15/32" TO Ø5"

## TSRE

One tool

| Model | 18 TSRE | 30 TSRE | 38 TSRE | 50 TSRE | 65 TSRE | 80 TSRE | 100 TSRE | 65 TSRE-RBE | 80 TSRE-RBE |
|---|---|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø11/16" (18 mm) × 16 ga (1.6 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-1/2" (38.1 mm) × 15 ga (1.8 mm) | Ø2" (50.8 mm) × 14 ga (2 mm) | Ø2-1/2" (65 mm) × 14 ga (2 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø4" (100 mm) × 11 ga (3 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.8 mm) |
| **Centreline radius (CLR)** | 0.4 – 3.1" (10 – 80 mm) | 0.6 – 4.7" (15 – 120 mm) | 1.0 – 5.9" (25 – 150 mm) | 1.2 – 8.7" (30 – 220 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.6 – 13.8" (40 – 350 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) |
| **Mandrel working length** | 3.9 ft · 47" (1,200 mm) | 7.2 ft · 87" (2,200 mm) | 7.2 ft · 87" (2,200 mm) | 8.2 ft · 98" (2,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) |
| **Tooling capacity**<br>stacked dies / radii | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| **Roll bending** | – | – | – | – | – | – | – | – | – |
| **Rear booster** | – | – | – | – | – | – | – | Servo | Servo |

## TDRE

Two stacked tools

| Model | 12 TDRE | 18 TDRE | 25 TDRE | 30 TDRE | 38 TDRE | 50 TDRE | 65 TDRE | 80 TDRE | 100 TDRE | 65 TDRE-RBE | 80 TDRE-RBE | 100 TDRE-RBE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Max. tube capacity**<br>OD × wall, mild steel | Ø15/32" (12 mm) × 20 ga (1 mm) | Ø11/16" (18 mm) × 16 ga (1.6 mm) | Ø1" (25.4 mm) × 16 ga (1.5 mm) | Ø1-3/16" (30 mm) × 14 ga (2 mm) | Ø1-1/2" (38.1 mm) × 15 ga (1.8 mm) | Ø2" (50.8 mm) × 14 ga (2 mm) | Ø2-1/2" (65 mm) × 14 ga (2 mm) | Ø3-1/8" (80 mm) × 12 ga (2.5 mm) | Ø4" (100 mm) × 11 ga (3 mm) | Ø2-1/2" (65 mm) × 12 ga (2.5 mm) | Ø3-1/8" (80 mm) × 12 ga (2.8 mm) | Ø4" (100 mm) × 10 ga (3.6 mm) |
| **Centreline radius (CLR)** | 0.5 – 2.2" (12 – 55 mm) | 0.4 – 3.1" (10 – 80 mm) | 0.6 – 4.7" (15 – 120 mm) | 0.6 – 4.7" (15 – 120 mm) | 1.0 – 5.9" (25 – 150 mm) | 1.2 – 8.7" (30 – 220 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) | 1.4 – 9.8" (35 – 250 mm) | 1.6 – 11.0" (40 – 280 mm) | 1.2 – 11.8" (30 – 300 mm) |
| **Mandrel working length** | 3.3 ft · 39" (1,000 mm) | 3.9 ft · 47" (1,200 mm) | 4.9 ft · 59" (1,500 mm) | 7.2 ft · 87" (2,200 mm) | 7.2 ft · 87" (2,200 mm) | 8.2 ft · 98" (2,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) | 8.2 ft · 98" (2,500 mm) | 11.5 ft · 138" (3,500 mm) | 11.5 ft · 138" (3,500 mm) |
| **Tooling capacity**<br>stacked dies / radii | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **Roll bending** | – | – | – | – | – | – | – | – | – | – | – | – |
| **Rear booster** | – | – | – | – | – | – | – | – | – | Servo | Servo | Servo |

Capacities are the manufacturer's ratings for round mild-steel tube, outside diameter × wall; CLR = centreline radius of the bend die. Mandrel length is the maximum tube length fed with the standard mandrel rod (the working length). Tooling capacity is the number of stacked bend dies / radii; "roll bending" models carry a roll-bending head for large radii; a rear booster (RBE servo, RBF / RBH / CBF hydraulic) pushes the tube from behind for thin-wall and tight-radius bends. Three-phase machines are supplied for 600 V with a CSA-approved panel.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

### INCLUDED WITH EVERY NOVOTECH MACHINE

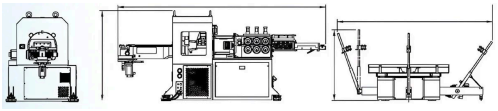
- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

WB 10-A · WB 14-A · 2 MODELS

# WB 10-A AND WB 14-A CNC WIRE BENDING MACHINES, Ø5/32" TO Ø9/16" WIRE FROM COIL



**MODEL** WB · CSM WB 10-A CNC WIRE BENDING MACHINE WITH DECOILER, STRAIGHTENER AND SERVO BENDING HEAD



WB wire bending line layout — decoiler, straightener, feeder and bending head



IPC control with 3D simulation of the wire bending machine

## 3D CNC from the coil · Ø5/32" – Ø9/16" · 1,000°/s · ±0.01°

A 3D CNC wire bender fed from the coil: the decoiler, hydraulic straightener and servo feeder bring the wire to a servo bending head that bends at up to 1,000°/s with ±0.01° repeatability, rotates the work-head 440°/s for the next plane and shifts between concentric tools, then a hydraulic cutter drops the finished part — baskets, racks, handles, hooks, springs, brackets and display frames in Ø5/32" to Ø3/8" (WB 10-A) or Ø1/4" to Ø9/16" (WB 14-A) wire. The customised work-head avoids interference on complex shapes; the IPC control with a 15" touch-screen simulates the part in 3D before the first wire is bent.

### STANDARD FEATURES

- ✓ Bending by servo motor — 1,000°/s (WB 10-A) / 600°/s (WB 14-A), ±0.01° repeatability
- ✓ Work-head rotation by servo motor, 440°/s, ±0.01°; tooling shift by servo motor, concentric tool design
- ✓ Feeding by servo motor at 164 ft/min, ±0.004"; straightening by hydraulics; uncoiling by AC motor
- ✓ Cutting by hydraulics; customised work-head eliminates interference
- ✓ IPC control with 15" touch-screen and 3D simulation
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

### OPTIONAL EQUIPMENT

- + Tooling to your wire and radii
- + Decoiler for large coils
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS

# WB 10-A AND WB 14-A CNC WIRE BENDING MACHINES, Ø5/32" TO Ø9/16" WIRE FROM COIL

## WB

CNC wire bending machines

| Model | WB 10-A | WB 14-A |
|---|---|---|
| **Bending capacity** wire OD | **Ø5/32" – 3/8" (4 – 10 mm)** | **Ø1/4" – 9/16" (6 – 14 mm)** |
| **Bending speed / repeatability** | **1,000°/s · ±0.01°** | **600°/s · ±0.01°** |
| **Feeding speed / repeatability** | 164 ft/min (50 m/min) · ±0.004" (±0.1 mm) | 164 ft/min (50 m/min) · ±0.004" (±0.1 mm) |
| **Work-head rotation speed / repeatability** | 440°/s · ±0.01° | 440°/s · ±0.01° |
| **Max. pressure** | 1,991 psi (140 kgf/cm$^2$) | 1,991 psi (140 kgf/cm$^2$) |
| **Total power** | 18.0 HP (13.4 kW) | 20.0 HP (14.9 kW) |
| **Dimensions L × W × H** | 136 × 51 × 68" (3,450 × 1,300 × 1,715 mm) | 136 × 51 × 68" (3,450 × 1,300 × 1,715 mm) |
| **Weight** | 5,291 lb (2,400 kg) | 5,291 lb (2,400 kg) |

Bending capacity is the wire diameter in mild steel; the WB 14-A bends the larger wire at 600°/s.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

### INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

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NT NovoTech MACHINE TOOLS Inc.


Specifications subject to change without notice. Imperial values are converted from manufacturer metric data and rounded: thin sheet in gauge (ga), plate and capacities in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment. 7-day money-back guarantee on in-stock machines only; financing available.